What's new

SAIR Open Math OS released

Terence Tao · 16 October 2026

https://terrytao.wordpress.com/2026/10/16/sair-open-math-os-released/

tao2typ · converted to Typst 2026-10-21 (UTC)

As announced a few weeks ago, the SAIR Open Math OS is now soft launched. It offers services and web interfaces similar to frontier AI models, though currently we are hosting existing open math models (DeepSeek V4.1 Flash, Kimi K3, and GLM 5.3 Flash) which are roughly a year or so behind the current frontier models in capability; so not able to perform "one-shot problem solving" of the type we have heard frequently about in recent months, but capable enough for many other useful mathematical tasks, such as medium-sized Lean formalization, or summarizing or discussing a mathematical argument or topic. Here is a typical screenshot:

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We are in the process of gathering partners to develop a more fine-tuned model for mathematics by post-training on an existing open model; see this link (and the associated Zulip channel) for more details.

One feature of our model is the emphasis on team accounts rather than individual accounts, in which members of a team see each other's prompts. We believe this could be a better format for using these tools for collaborative projects.

We are hosting these models using donated compute, and are in the process of granting free accounts to requesters from academic institutions. We may need to adjust the allocation of compute depending on demand and on future donations, but we will try to make the models available to as many users as possible.

This is still a very early version of the model; we accelerated our timeline for first release due to the increased demand for open models from the community. We will continue developing the model and accompanying features and welcome feedback, for instance at the Zulip channel.

Comments (17)

ryeguy10 · 16 October 2026

Excellent! Probably this will not be useful in the near future. But I do suspect the current LLM boom to slow down at some point at least as far as mathematics is concerned, and then open source models will catch up.

· 17 October 2026

Can I ask why one should expect this? When you say slow down you mean LLMs in general will not progress as quickly, or that the gap between open source and the current frontier grow smaller? And why would something with unlimited RLVR capability slow down?

ryeguy10 · 18 October 2026

I mean, there are only so many orders of magnitude in the universe, so the current LLM boom will slow down at some point. It's been working for a few years. Where it stops no one knows.

· 16 October 2026

Do you eventually plan on pretraining a model from scratch?

Terence Tao · 18 October 2026

Pretraining is a very long-term goal for our foundation, it would require significantly more compute than we plan to fundraise for. Our near-term goal is to focus on post-training an existing open-source model with data of a mathematical nature to fine tune its capabilities at mathematical tasks (which will include some problem-solving tasks, although this will not be our primary focus).

shacharlovett · 16 October 2026

Very nice, I want to experiment with it. What is the process of getting access?

dpthurst · 16 October 2026

One question: like most mathematicians, I have multiple projects that I work on with a variety of subsets of people. (Typically I'll be working in a general area with a group of 3 to 5 people, and the collaboration would cover multiple papers.) Is the model to apply for a separate team account for each of these collaborations?

Currently I just applied for an account for myself, for the few things that I'm not (currently) working with anyone else on.

Terence Tao · 18 October 2026

One would need separate team accounts if one wanted to avoid sharing the prompts from one team with another. We are still in the process of deciding what limits to place on account creation to prevent a small number of users absorbing all the available donated compute that we have, in particular we may place a cap on how many accounts a given institution or user can have, but we would like to first collect some usage data before finalizing our policies in this regard.

· 16 October 2026

Why not offer a downloadable version that can be run locally instead of having it on the cloud?

dpthurst · 17 October 2026

The hardware requirements for at least Kimi K3 are extremely prohibitive; running that model locally is not going to be an option for almost anyone.

Titi · 17 October 2026

Why not allowing for local use ? Seems like a strange take...

· 17 October 2026

Does it say somewhere that local use will not be allowed?

Anonymous · 17 October 2026

I suppose because this initiative is very new and you cannot do everything at once. They say that the models will be open weight and the inference code open source::

https://www.openmathmodel.org/initiative/

Terence Coelho · 17 October 2026

I'm a little concerned that there isn't any mention of safety/alignment. I expect training for a combination of math capability and the ability to automate AI research to boost math capabilities of the next model to be the kind of target that is natural to shoot for with a small team that want to catch up to the frontier in math capabilities, but can go very very wrong without extremely careful safety work.

Terence Tao · 20 October 2026

We are still in the process of setting up the various subcommittees that will handle various aspects of the training, data gathering, evaluations, etc., of our future model but we definitely plan to make various safety assessments as part of our evaluation. (Hopefully experts in these areas will offer their services to that subcommittee.)

· 18 October 2026

Currently, everyone I know in my field works in fluid teams: 1 to 5 colleagues collaborating on a *project* until it yields a paper. Then team boundaries shift for the next project.

What would better suit many mathematicians is having individual accounts alongside team projects. Sharing prompts, conversations, and outputs within a project is extremely useful, but this needs to stay at the project level to preserve our fluid, bottom-up workflow.

It is quite unsettling that the plan here seems so different—expecting researchers to commit to a fixed team, with a single person applying for everyone and becoming the official "boss." It feels top-down and disturbing. Is it understood that our relatively horizontal model—which accounts for 90% of the joy at work—is over? I don't see why it should be discouraged, nor why the arrival of AI should change that (and I've seen AI as a good thing for math so far).

Nurma · 18 October 2026

congrats on the release! we'd love to help improve autoformalization with our rl environments